Theme 1.5 — Settlements & Service Provision

Case Study: Lisbon Metropolitan Region, Portugal

This pack covers **Cambridge IGCSE 0460 syllabus objective 1.5**: settlement patterns (dispersed, linear, nucleated), the factors that influence site and situation, settlement hierarchy, high- and low-order services, sphere of influence and threshold population. All examples are drawn from the **Lisbon Metropolitan Region** — our local case study — so you can recall specific named places, features and services under exam conditions.

Key Terms

Site — the actual land on which a settlement is built (e.g. a hilltop, river crossing, flat plain).
Situation — the position of a settlement relative to other features (other towns, rivers, roads, the coast).
Settlement pattern — the shape a settlement takes when viewed from above — dispersed, linear or nucleated.
Hierarchy — the ranking of settlements by size, population and the services they provide.
Sphere of influence — the area served by a settlement or service — the area from which it draws customers.
Threshold population — the minimum number of people needed to support a service so it makes a profit.
High-order service — a specialist service (e.g. department store, hospital) with a large sphere of influence.
Low-order service — a convenience service (e.g. newsagent, primary school) with a small sphere of influence.

1. Settlement Patterns — Dispersed, Linear, Nucleated

Óbidos — a classic nucleated village clustered inside medieval walls around the central church.

Estoril coast — linear settlement strung along the coastal road and railway between Lisbon and Cascais.

Alentejo — dispersed farmsteads (montes) scattered widely across flat agricultural land.

Sintra town centre — a small nucleated settlement below the Moorish castle hill.

Nucleated settlements cluster tightly around a central point such as a church, square or crossroads. In the Lisbon region, **Óbidos** is the textbook example: houses are squeezed inside a ring of medieval walls, with the Igreja de Santa Maria at the centre. Nucleation developed here for **defence** (hilltop walled site) and **shared services** (well, church, market). **Sintra** town and **Colares** village also show nucleated patterns, clustered around their historic cores.

Linear settlements form long, thin strips along a transport route, coastline or valley. The **Estoril–Cascais coastline** is a textbook linear pattern: development follows the coastal railway (opened 1889) and the Marginal road, creating an almost unbroken ribbon of housing, hotels and shops from Oeiras to Cascais. Linear settlements spread because access is easiest *along* the transport line, and the sea limits expansion inland on one side.

Dispersed settlements consist of isolated farms or houses scattered across the landscape with no central focus. Across the border of the Lisbon region, the **Alentejo** plains show a dispersed pattern of single *montes* (large isolated farmsteads) separated by hundreds of metres of open fields. Dispersal occurs where land is flat and agriculturally productive, where farms are very large, and where water is available from individual wells rather than a shared source.

Pattern	Typical site factors	Lisbon Region example
Nucleated	Defensive hilltop; single water source; fertile land all around; need for shared facilities	Óbidos (walled hilltop); Sintra (below castle); Colares (around church and square)
Linear	Coastline, valley floor, river, road, railway; narrow strip of flat land	Estoril–Cascais ribbon along the coastal railway and Marginal road
Dispersed	Flat fertile land; large farm sizes; water available everywhere (individual wells); safe region	Alentejo montes (just south of the Lisbon region); scattered farms in rural Mafra

Exam Question 1 [2 marks]

Define the term **nucleated settlement**.

✓ Mark scheme — Q1 (2 marks)

A settlement where buildings are **clustered** / grouped tightly together (1) around a **central point** such as a church, square or crossroads (1). *Accept: compact, concentrated, not spread out.*

Exam Question 2 [4 marks]

Using a named example, describe the pattern of a **linear settlement**.

✓ Mark scheme — Q2 (4 marks)

1 mark for a **named example** (e.g. Estoril / Cascais coast). Up to 3 marks for description: buildings stretched in a **long thin line** (1); follow a **transport route** — coastal railway / Marginal road (1); development is only a few streets **wide** but many kilometres **long** (1); the sea / coast **limits growth** on one side (1).

2. Site & Situation — Why Lisbon Grew Where It Did

São Jorge Castle — the original defensive site on a steep hilltop overlooking the Tagus.

The Tagus estuary — Europe's largest natural harbour, sheltered from Atlantic storms.

Lisbon today — the modern city spreads down from its seven hills to the river.

Óbidos — a smaller defensive hilltop site, also walled for protection.

The **site** of a settlement is the physical land it is built on; the **situation** is how that site sits in relation to its surroundings. Lisbon's original site was the **castle hill (São Jorge)** — a steep 110 m hilltop that was **easy to defend**, had a **commanding view** over approaching ships, and offered a reliable spring for water. Settlements needed all three — defence, view, water — to survive in medieval times.

Lisbon's **situation** is what turned it from a hilltop fort into a capital city. It sits at the mouth of the **River Tagus**, whose estuary forms one of Europe's largest **natural harbours**: sheltered from Atlantic storms, deep enough for ocean-going ships, and with flat foreshore land for docks and warehouses. From the 15th century this situation drove the Age of Discovery — fleets sailed from here to Africa, India and Brazil. Today the same situation supports the Port

of Lisbon, cruise terminal and Humberto Delgado Airport.

	Site (the land itself)	Situation (the position relative to…)
Lisbon	Seven steep hills, hilltop with spring water, defensive position above the river	Mouth of the Tagus; natural harbour; Atlantic coast; centre of Portugal
Óbidos	Small limestone hill with a flat summit, surrounded by fertile farmland	80 km N of Lisbon on a historic road; near the Atlantic but inland enough for safety
Cascais	Sheltered bay on the coast, flat land behind for expansion	Mouth of the Tagus; terminus of the Lisbon–Cascais railway; resort hinterland

Exam Question 3 [3 marks]

Explain the difference between the **site** and the **situation** of a settlement.

✓ Mark scheme — Q3 (3 marks)

Site = the **actual land / physical ground** a settlement is built on (1). Situation = the **position of the settlement relative to surrounding features** such as other towns, rivers, roads or coast (1). Credit 1 mark for a clear contrast or example (e.g. Lisbon's *site* is a hilltop with a spring; its *situation* is the mouth of the Tagus).

3. Settlement Hierarchy & Service Provision

Colares village — small local restaurant and shop: low-order services only.

Sintra town — cafés, banks, pharmacies: mid-order services for a wider area.

Cascais — Mercado da Vila offers specialist food, restaurants and tourist services.

Centro Colombo, Lisbon — department stores, cinemas, 340+ shops: highest-order services.

A **settlement hierarchy** ranks places by size and the services they provide. Small settlements have only **low-order (convenience)** services that people use often and will not travel far for. Large settlements also offer **high-order (comparison)** services that are used occasionally and that people will travel considerable distances to reach. The Lisbon Metropolitan Region shows every level of this hierarchy within an hour's travel.

Level	Settlement type	Lisbon Region example	Typical services
1	Isolated farm / hamlet	Montes in rural Mafra; farmhouses in Colares vineyards	None — residents travel to nearest village
2	Village	Colares (~7,500 people)	Primary school, café, small grocery, pharmacy, church
3	Small town	Sintra (~30,000 in urban area)	Secondary school, supermarket, bank, medical centre, post office

Level	Settlement type	Lisbon Region example	Typical services
4	Large town	Cascais (~35,000 in urban parish; 200,000+ municipality)	Hospital, shopping streets, markets, cinemas, train terminus
5	City / capital	Lisbon (~550,000 city; 2.9 million metro)	Universities, major hospitals, Colombo & Vasco da Gama malls, international airport, government

Sphere of influence is the area a service draws customers from. A **village bakery in Colares** has a sphere of influence of perhaps 2 km — only locals use it. A **hospital in central Lisbon** may have a sphere of influence covering the entire Greater Lisbon region (2.9 million people) because specialist treatments are not available elsewhere. High-order services always have larger spheres of influence than low-order services.

Threshold population is the minimum number of people needed to make a service profitable. A corner café might need only 300 regular customers; a supermarket needs around 3,000; a hospital needs 300,000+; a major department store like **El Corte Inglés** in Lisbon needs a threshold of around half a million people — which is why it is only found in the largest city. This explains why Colares cannot support a department store, but Lisbon can.

Exam Question 4 [4 marks]

With reference to a named city or region, explain why some services are only found in the largest settlements.

✓ Mark scheme — Q4 (4 marks)

1 mark for **named example** (e.g. Lisbon / Centro Colombo). Up to 3 marks for explanation: some services need a high **threshold population** to make a profit (1); only large settlements have enough potential customers to meet this threshold (1); they have a **large sphere of influence** because people will travel far for specialist / comparison goods (1); good transport links into the city support this (1). *Specific examples expected: department store / major hospital / university / international airport.*

4. Exam Practice — Extended Response

The 7-mark question at the end of Theme 1 usually asks you to **use a named case study** to explain settlement patterns, hierarchy or services. You must include **specific places, figures and examples** from the Lisbon Metropolitan Region. Use the **PEEL framework** below to structure each paragraph and aim for **3 developed points**.

Exam Question 5 [7 marks]

Using a named case study, explain how **settlement hierarchy** and **sphere of influence** affect the services available in different settlements. *Use specific named places and examples in your answer.*

PEEL Framework — use this for the 7-mark answer

Prompt: Write three PEEL paragraphs — one for each level of the hierarchy (e.g. Colares village → Sintra / Cascais town → Lisbon city).

Point — State your main idea clearly.
Evidence — Use specific facts, figures or named places from Lisbon.
Explain — Link cause to effect using geographical reasoning.
Link — Connect back to the question and make a judgement.

✓ Mark scheme — Q5 (7 marks)

Level 3 (6–7 marks): Detailed answer with **named case study** (Lisbon Metropolitan Region) and **specific examples** at three different hierarchy levels. Clear link between sphere of influence / threshold and the services available. Correct use of geographical terminology (low-order, high-order, threshold, sphere of influence).
Level 2 (4–5 marks): Some named examples but may only cover two hierarchy levels. Partial link between threshold and service availability. Some terminology.
Level 1 (1–3 marks): General statements without named examples. Simple description of service differences with little explanation. Limited terminology.

Model content points (credit any three developed): (i) **Colares village** (~7,500 people) has only **low-order services** — primary school, small grocery, pharmacy — because its **threshold population** is too small to support specialist shops, and its **sphere of influence** is only a few km. (ii) **Sintra town** (~30,000) offers **mid-order services** — supermarkets, banks, medical centre — because its larger threshold and wider sphere of influence (surrounding villages travel in for weekly shopping) supports them. (iii) **Lisbon** (2.9 million metro) provides **high-order services** — **Centro Colombo** with 340+ shops, major hospitals, universities, international airport — because only a capital city has a large enough threshold population and wide enough sphere of influence (national and international) to support them.

5. Quick Recall — Must-Know Facts for Lisbon

Fact category	Number / detail to memorise
Lisbon city population	~550,000 (city proper)
Lisbon Metropolitan Area population	~2.9 million (largest in Portugal)
Lisbon's original site	Hilltop of São Jorge Castle (~110 m), with spring water
Lisbon's situation	Mouth of the River Tagus — largest natural harbour in Europe
Óbidos population	~3,100 — walled nucleated hilltop village
Sintra urban population	~30,000 — small nucleated town below Moorish castle
Cascais municipality	~215,000 — resort town, terminus of coastal railway
Colares population	~7,500 — nucleated village in the Sintra hills
Centro Colombo	340+ shops, ~26 million visitors/year — high-order retail

Fact category	Number / detail to memorise
Estoril–Cascais rail line	Opened 1889, drove linear development along the coast
Sphere of influence — village shop	~2 km (Colares grocery)
Sphere of influence — capital hospital	Entire metro area: ~2.9 million

Summary — The Big Picture

Settlement geography explains **where people live and why**. Patterns (dispersed, linear, nucleated) reflect the **physical environment** — flat farmland, coastlines, defensive hills. The **site** of a place is the land it sits on; the **situation** is how that place relates to its surroundings — and situation usually matters more for long-term growth, which is why Lisbon (mouth of the Tagus) became a capital while equally defensive hilltop villages like Óbidos stayed small. The **hierarchy** from hamlet → village → town → city reflects **threshold population** and **sphere of influence**: only large populations can support specialist high-order services like department stores, hospitals and universities, which is why these concentrate in Lisbon while Colares has only a primary school and grocery.

! EXAM ALERT

In the exam, **always name real places** from Lisbon Metropolitan Region — Óbidos, Estoril, Sintra, Colares, Cascais, São Jorge Castle, the Tagus estuary, Centro Colombo. Vague answers ("a big city", "a small village") lose marks. For 7-mark questions, use **PEEL** and cover **three hierarchy levels** with specific services, numbers and distances. Check the command word — *describe* = say what you see; *explain* = give reasons why; *compare* = highlight similarities AND differences.

Revision Checklist — tick when confident

- [ ] I can define **site, situation, hierarchy, sphere of influence** and **threshold population**.
- [ ] I can describe and give a named Lisbon-region example of **dispersed, linear** and **nucleated** patterns.
- [ ] I can explain why Lisbon's **site** (São Jorge hill) and **situation** (mouth of the Tagus) made it grow.
- [ ] I can place **Colares, Sintra, Cascais** and **Lisbon** on a 5-level hierarchy and name their services.
- [ ] I can explain why **Centro Colombo** is in Lisbon and not in Colares, using threshold and sphere of influence.
- [ ] I can write a **PEEL** 7-mark answer covering three different hierarchy levels with specific named examples.